

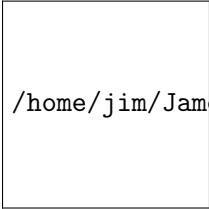
Section 2.4: Permutations and Combinations

Finite Mathematics MTH 107 Fall 2014

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Mars Hill University

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2014



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- ex. Choose 2 out of 10 people for president and vice president?
- ex. Choose 3 out of 6 vegetable toppings. How many different pizzas are possible?

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Example 2.4.1.

5 students are giving presentations. How many different lineups are possible?

- $5 \times 4 \times 3 \times 2 \times 1 =$

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What if there is only time for 3 out of the 5 students to give a presentation?

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$$5 \times 4 \times 3 = 60$$

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Recall: $0! = 1$

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- No replacement, order matters, and selected from 1 category.

Combinations

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 $A = \{a, b, c\}$?

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Removing the repeats, $\frac{6}{2} = 3$.

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Removing the repeats, $\frac{6}{2} = 3$.

$$\left(\frac{3!}{(3-2)!2!} \right)$$

Combinations

Combination Formula

The number of distinct *combinations* of r items selected without replacement from a pool of n items ($r \leq n$),

$${}_nC_r = \frac{n!}{(n-r)!r!}$$

Combinations are used whenever one or more items are selected (without replacement) from a category and the order of selection is not important.

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For each pizza with 3 of the same toppings, there are $3!$ ways to place the order.

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ex. $\{Pep_1, Ham, Pep_2\} = \{Ham, Pep_1, Pep_2\} =$
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So, there are $\frac{8!}{(8-3)!} \frac{1}{3!} = 56$ different pizzas.

Permutation or Combination?

Problem 2.4.1. Find the indicated value.

11 people try out for a baseball team, but there are only 9 spots.
How many different teams are possible?

Permutation or Combination?

Problem 2.4.2. Find the indicated value.

A committee of 3 students must be selected for student council from a pool of 6. How many different student councils are possible?

Permutation or Combination?

Problem 2.4.3. Find the indicated value.

A committee of 3 students must be selected for student council from a pool of 6. One will be president, one vice president, and one will be janitor. How many different student councils are possible?

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Problem 2.4.4. Find the indicated value.

A committee of 3 students must be selected for student council from a pool of 6. One will be president and two will be vice presidents. How many different student councils are possible?

Permutation or Combination?

Problem 2.4.5. Find the indicated value

A baseball league has 6 teams. If every team must play every other team once in the first round of league play, how many games must be played?

When there is more than one category

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A pool of 6 seniors, 5 juniors, 4 sophomores, and 3 freshmen must select a committee of 4. How many committees are possible if the committee must contain

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A pool of 6 seniors, 5 juniors, 4 sophomores, and 3 freshmen must select a committee of 4. How many committees are possible if the committee must contain

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We have 4 categories

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$${}_{18}C_4 = \frac{18!}{(18-4)!4!}$$

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$$\underline{6}C_2 \times \underline{5}C_2$$

Distinguishable Permutations

Example 2.4.8.

Find the number of *distinguishable* permutations of the letters of the word “TRAIN”

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Example 2.4.9.

Find the number of *distinguishable* permutations of the letters of the word “TENNESSEE”

There are $9!$ ways to arrange 9 items, but “TENNESSE^{blue}E^{red}” is not distinguishable from “TENNESSE^{red}E^{blue}”.

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- There are $4!$ ways to arrange 4 “E”’s,
- $2!$ ways to arrange 2 “N”’s,
- and $2!$ ways to arrange 2 “S”’s.

So there are $\frac{9!}{4!2!2!} = \frac{9 \cdot 8 \cdot 7 \cdot 6 \cdot 5}{4} = 3,780$ *distinguishable* arrangements.

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How many five-card hands consisting of all hearts are possible?

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Two categories, “ways to get a suit” and “ways to get one suit”, $\underline{4} \times \underline{{}_{13}C_4}$.

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How many five-card hands consisting of 3 kings and 2 queens are possible?

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